Effect of Physical Activity on Heart Rate

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TJ Math 3

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Part 1: Experimental Design

Introduction:

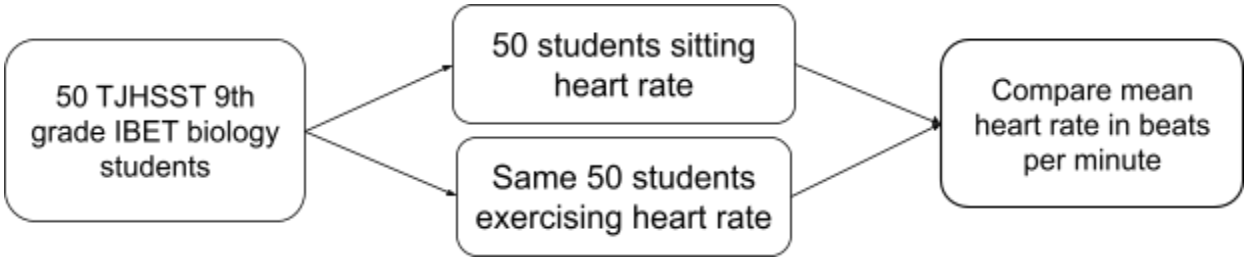
Have you ever felt out of breath or exhausted, or heard the drumming noise of your heart pounding quickly and loudly in your ears after performing a high intensity exercise or physical activity for a long period of time? We hope to find out more about the relationship between performing physical activity and the resulting heart rate after doing an exercise and provide insights on managing heart rate while exercising to not feel out of breath. We believe that if the intensity or movements required for an exercise increases and someone performs it for a consistent amount of time and then measures their heart rate afterwards, then their heart rate will increase as a result of the increased activity. The muscles demand more oxygen while exercising, which is transported by an increased blood flow from the heart beating faster to be able to get enough oxygen to the muscles.

Experimental Design Table:

Title: Effect of Physical Activity on Heart Rate
Hypothesis: If someone performs a physical activity with higher intensity that needs more movements and measures their heart rate afterwards, then their heart rate in beats per minute after performing the activity will increase
Experimental Units: 467 Thomas Jefferson High School for Science and Technology

9th grade IBET biology students		
Randomization: Simple random sample of 50 students collected from the population		
Independent Variable (IV): Whether the student was sitting or exercising before taking the heart rate (categorical binary)		
Treatment Groups: (matched pairs)	Sitting (control)	Exercise (treatment)
Independent Trials:	50	50
Dependent Variable (DV): Heart rate after performing physical activity in beats per minute (quantitative)		
Constants: Duration of performing physical activity, method of measuring heart rate, and method beats per minute is calculated.		

Experimental Design Diagram:



Methods:

All 467 9th grade IBET biology students in Thomas Jefferson High School for Science and Technology were told to measure their heart rate while they were sitting by either using pulse oximeters or counting the pulses on their wrist in one minute, measured by a 60 second timer. The same procedure was repeated 3 times in total per student, and each student recorded the average heart rate of their 3 trials in beats per minute. The entire process was repeated again, where the students had to measure their heart rate after doing an exercise of their choice while standing. After every student recorded their data, a mean heart rate was calculated for the sample of 50 students' mean sitting heart rate and a mean heart rate was calculated for the same 50 students' mean heart rate after exercising for 30 seconds.

Part 2: Data and Description

Data:

Sitting, exercising, and heart rate difference data are all calculated in beats per minute (BPM), so a value of 67 means 67 BPM.

Student Number	Mean Sitting Heart Rate	Mean Exercising Heart Rate	Difference	Student Number	Mean Sitting Heart Rate	Mean Exercising Heart Rate	Difference
1	79	110	31	26	72	108	36
2	91	111	20	27	48.2	55.9	7.7
3	80	146.7	66.7	28	80.3	129.7	49.4
4	92.3	129	36.7	29	72.3	136	63.7
5	72.3	105	32.7	30	58.3	85.3	27
6	90	115	25	31	72.6	92.6	20
7	57.7	94.7	37	32	61	154	93
8	70	108	38	33	72.7	114.7	42
9	55	76	21	34	85	116	31
10	93.3	126.7	33.4	35	105	130	25
11	65.3	97.3	32	36	77.7	132.3	54.6
12	74	143.7	69.7	37	84	137	53
13	73.3	114.7	41.4	38	69	89	20
14	64.3	108	43.7	39	72	107	35
15	65.3	128	62.7	40	68	108	40
16	79	137	58	41	78.6	131.6	53
17	54	147.3	93.3	42	66	84	18
18	86.6	120	33.4	43	62.7	82.7	20
19	72	117	45	44	76	104	28
20	75	124.6	49.6	45	73	130	57
21	77.3	125.3	48	46	72.5	116	43.5
22	94.5	140	45.5	47	65	85.3	20.3
23	40	105	65	48	56	101.3	45.3
24	77	128	51	49	69.3	104	34.7
25	99.8	110.7	10.9	50	74.7	126.7	52

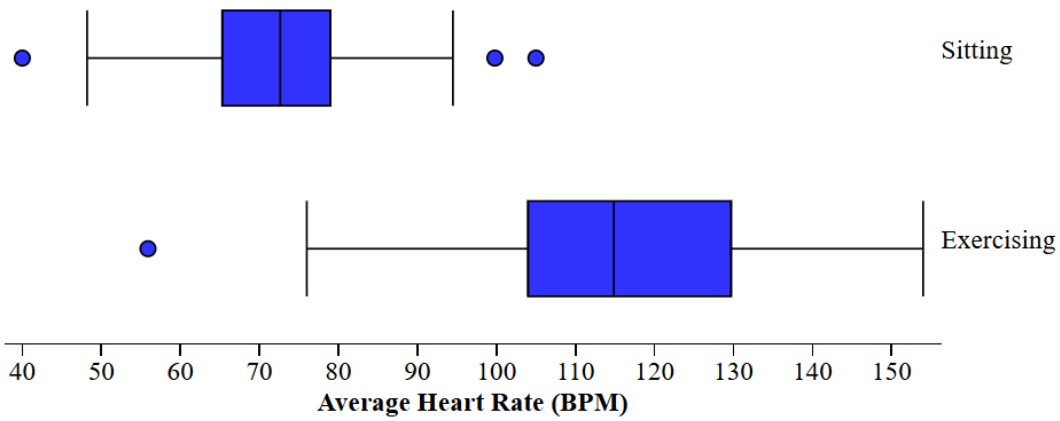
Data Analysis:

Size (n)	Mean	SD	Min	Q1	Median	Q3	Max
50	73.398	12.839	40	65.3	72.65	79	105

Average Sitting Heart Rate Sample Statistics

Size (n)	Mean	SD	Min	Q1	Median	Q3	Max
50	114.596	20.37	55.9	104	114.85	129.7	154

Average Exercising Heart Rate Sample Statistics



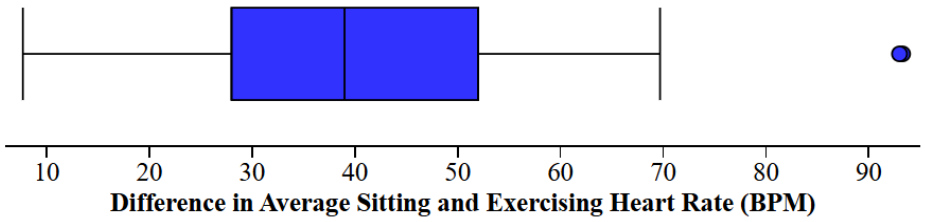
Average Sitting and Exercising Heart Rates Comparative Boxplots

The distributions of 9th grade IBET biology students at Thomas Jefferson High School for Science and Technology's average sitting heart rates and the same students' average exercising heart rates in beats per minute (BPM) are both roughly mound shaped. Based on the calculations (65.3(Q1)-1.5*13.7(IQR), 79(Q3)+1.5*13.7(IQR)) there are outliers for 9th grade IBET biology students' average sitting heart rate and based on the calculations (104(Q1)-1.5*25.7(IQR), 129.7(Q3)+1.5*25.7(IQR)) there is an outlier for 9th grade IBET biology students' average exercising heart rate. The medians are 72.65 BPM for students' average sitting heart rate and 114.85 BPM for

students' average exercising heart rate, and the IQR's are 13.7 BPM for students' average sitting heart rate and 25.7 BPM for students' average exercising heart rate. A typical 9th grade IBET biology student at Thomas Jefferson High School for Science and Technology tends to have a sitting heart rate of around 72.65 BPM with a smaller IQR of 13.7 BPM, compared to their typical exercising heart rate of around 114.85 BPM, which is 42.2 BPM higher, and a higher IQR of 25.7 BPM.

Size	Mean	SD	Min	Q1	Median	Q3	Max
50	41.198	18.411	7.7	28	39	52	93.3

Average Exercising Heart Rate Sample Statistics



Difference in Average Sitting and Exercising Heart Rate Boxplot

Since we're using a matched pairs design, we need to consider the distribution of all the pairs' differences. The distribution of the difference in the average sitting and exercising heart rates of 9th grade IBET biology students at Thomas Jefferson High School for Science and Technology in beats per minute (BPM) is right-skewed. Based on the calculations (28(Q1)-1.5*24(IQR), 52(Q3)+1.5*24(IQR)) there are outliers. The median is 39 BPM and the IQR is 24 BPM. The exercising heart rate of a typical 9th grade IBET biology student at Thomas Jefferson High School for Science and Technology tends to be 39 BPM greater than their average sitting heart rate, with an IQR of 24 BPM.

Part 3: Statistical Inference and Conclusions

Further Data Analysis

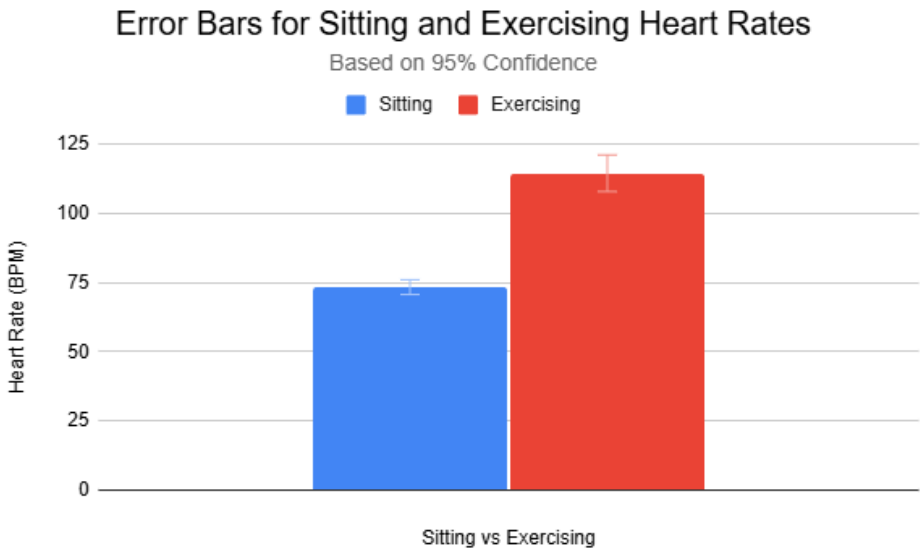
For difference in students' heart rate in BPM between the two activities:

	1 SD from Mean	2 SD from Mean	3 SD from Mean
Interval	(22.787, 59.411)	(4.376, 77.822)	(-14.035, 96.233)
Number of Students	34	48	50
Percent of Students	68%	96%	100%
Empirical Rule	68%	95%	99.7%

The distribution of the difference in the students' heart rate in BPM between sitting and exercising is roughly mound-shaped and the percent of observations in each interval (1, 2, 3 SD from the mean) are very close to the Empirical Rule percentages. It is plausible that the heart rate difference distribution came from a normally distributed population.

Inference

Error Bar Graph:



The error bars for the 95% confidence interval are not close to overlapping at all, so this is further evidence that there is a significant difference in the students' mean heart rates between sitting and exercising.

Confidence Interval:

(35.966, 46.43)

We are 95% confident that the population mean of a student's exercising heart rate in BPM would be 35.966 to 46.43 more than the same student's sitting heart rate in BPM.

Define Parameters:

The population mean of the difference in the students' heart rate in BPM between sitting and exercising is μ_d.

Hypotheses:

H_0: $\mu_d = 0$

H_a: $\mu_d > 0$

Technical Conditions:

1. The 50 students from the sample were randomly chosen out of all of the 9th grade IBET biology students at TJHSST.
2. The sample size is larger than 30.
3. The sample is not independent because the sample size is greater than 10% of the population (50 > 46.7), however, we can infer that the population is roughly normally distributed based on the further data analysis section.

Test Statistic:

$t = \frac{\overline{x_d}}{s_d/\sqrt{n}} = \frac{41.198}{18.411/\sqrt{50}}$ = 15.823

P-Value:

p-value = 3.418×10^{-21} = 0.000000000000000000003418.

Test Conclusion:

We have evidence to reject the null hypothesis at the $\alpha = 0.05$ level since the p-value = $3.417 \times 10^{-21} < 0.05$. We have highly significant evidence that the mean difference between a student's exercising and sitting heart rate in BPM is greater than zero, meaning a student's exercising heart rate tends to be higher than the same student's mean sitting heart rate.

Conclusion

The purpose of this experiment was to find out more about the relationship between performing physical activity and the resulting heart rate after doing an exercise and provide insights on managing heart rate while exercising to not feel out of breath. Based on the boxplot of the difference in students' sitting and exercising heart rates in BPM, students' exercising heart rates tended to be higher than their sitting heart rates. Using the p-value from a one-sample matched pair t-test, we found highly significant evidence to reject the null hypothesis that there was no difference between the two heart rates. A plausible explanation for this finding is that muscles are used more when exercising compared to sitting, increasing the body's overall demand for oxygen. To match this demand, heart rate increases to pump oxygen faster throughout the entire body. Possible experimental design improvements include: everyone using pulse oximeters to

measure their heart rate instead of only some using them and others measuring their pulses manually, making everyone do the same kind of exercise, as people could choose what kind of physical activity they did for their exercising heart rate, and adding blocking between male and female for comparing the heart rate results. With these improvements incorporated into the experiment, we could get more accurate and reliable experiment results to further back our hypothesis and apply it to a broader audience.